① Data Processing 1.7.

(1) What is "Data Processing" in the context of Online Experiments.

In online A/B experiment, Data Processing refers to the initial steps after data is logged from user interactions, where raw logs are:

- Cleaned
- Parsed
- Filtered
- Aggregated by unit of analysis (e.g. per-user, per-session, per-day)

⇒ Purpose: Prepare data so it's accurate, consistent, and analysis-ready.

Real-World Analogy

Imagine you're running a pizza shop and you want to see if changing the cheese improves customer satisfaction.

- Raw logs = every customer's order (some duplicated, messy, incomplete)
- Data processing = removing duplicate orders, filtering incomplete one, grouping by customer ID.
- Only then can you analyze: "Did customers who got new cheese rate higher on average?"

Why it Matters

If you don't properly process the logs:

- Duplicates inflate metrics
- Missing data causes bias
- Wrong keys (e.g. using session instead of user ID) corrupts aggregation.

This links directly to Unit 14:

> # "Randomization Unit vs. Analysis Unit" — If you randomize at user level but aggregate at session level, your results are misleading.

(2) Key Concept in Data Processing

1. ETL (Extract, Transform, Load).

1. ETL (Extract, Transform, Load)
   - Extract raw logs from data sources (e.g. click logs, server logs)
   - Transform by cleaning, deduplication, fixing missing values
   - Load into the experiment data warehouse, grouped per analysis unit.

2. Filtering:
   - Remove bots, test traffic, corrupt records.

3. Joining Data:
   - Combine logs from different systems (e.g. client logs + backend orders).

4. Time Alignment
   - Ensure timestamp align with the ramp period and experiment start.

5. User Bucketing check
   - Ensure each user is always in the same experimental bucket (A or B).

(3) Underlying Math Concept: Aggregation Over Randomized Units.

Say you're measuring average revenue per user:

$$\text{Revenue}_{avg} = \frac{1}{n} \sum_{i=1}^{n} \text{Revenue}_i$$

If you don't process correctly:

- A single user might be counted twice
- A bot might have $1000 in revenue, skewing the average.

* Bias & variance increase from dirty data, leading to invalid statistical inference.

(4) Overlapping Concepts from Other Units.

Topic	Link
Unit 13 - Instrumentation	→ Garbage in = garbage out - instrumentation affects raw data quality.
Unit 14 - Randomization Unit	→ You must process data according to the correct unit of analysis (user/session)
Unit 15 - Ramping	→ During ramping, logs must be filtered by ramp stage + time window.
Unit 17 - Computation	→ Processing is the precursor to metrics computation and statistical analysis.

(5) Summary

Aspect	Description.
Goal	Prepare raw logs for valid experiment analysis.
Key steps	Clean, deduplicate, filter, align, aggregate.
Math	Bias - Pre-a aggregation over correct unit.
Tools	SQL, pandas, Spark etc.
If Ignored	Biased metrics, invalid conclusion.

dhanvadarje.github.io

② Data Computation 178

~~② What is "~~

① What is "Data Computation" in Online Experiments?

Once data is processed (cleaned, filtered, and aggregated), the next step is Data Computation — where we compute experiment metrics and perform statistical calculations to evaluate the effect of the experiment.

> It transforms processed logs into key metrics: click-through rates, conversion rates, average revenue, etc. with confidence intervals and statistical significance.

Real-World Analogy: Pizza A/B Test Continues

You previously cleaned your pizza order data (Data Processing). Now, you're asking:

> 🍕 "Did cheese version B increase customer satisfaction score?"

You now:

- Compute average satisfaction for group A and B.
- Calculate difference, confidence interval, p-value and determine if the change is statistically significant.

That is data computation.

(2) Main Computation Tasks:

1. Metric Computation

Compute core metrics like:

- Mean revenue / user
- Click-through rate (CTR)
- Conversion rate.

Example

$$CTR = \frac{\text{Clicks}}{\text{Views}}$$

2. Delta (Difference Between Variants)

Compare metrics between treatment (T) and control (C):

$$\Delta = \bar{X}_T - \bar{X}_C$$

3 Standard Error (SE) and Confidence Interval (CI)

used to quantify uncertainty:

$$SE = \sqrt{\frac{S_T^2}{n_T} + \frac{S_C^2}{n_C}}$$

$$CI = \Delta \pm 2 \cdot SE$$

4/ Statistical Significance

Calculate p-values to check if observed difference is due to change.

5/ Guardrail Metrics

Compute metrics that ensures no ~~se~~ negative side-effect (e.g. latency, errors) Important for trustworthy experiment.

Mathematical foundation (Example)

Let's say:

- Control group (C) average revenue: $25 (n=500, std=10).
- Treatment group (T) average revenue: $27 (n=500, std=12)

We compute:

Difference:

$$\Delta = 27 - 25 = 2.$$

SE:

$$SE = \sqrt{\frac{10^2}{600} + \frac{12^2}{500}} \approx 0.714.$$

95% CI:

$$CI = 2 \pm 1.96 \times 0.714 = (0.6, 3.4).$$

③ Related/Overlapping Topics.

Topic	Overlap.

unit5: Metrics Definitions → Metrics must be well-defined before computing.

unit6: Statistical Analysis → Computation of SE, p-values, confidence interval.

unit17: Summary & Viz → Computed data feeds directly into dashboard or notebooks.

unit20: Statistical Power → How computation helps detect real effect, not noise.

drawadarie ghising

(9) Summary.

Aspect	Descriptions
Purpose	Compute final metrics, compare variants, access impact.
Tasks	Mean, delta, standard error, CI, p-value
Math	Differences of means, Bernoulli variance, z-test.
Output	Results that determine if your experiment "won".
Risk	Wrong computation = misleading conclusion.

③ Results Summary and Visualization 180.

① What is "Results Summary and Visualization"?

After data is:

- ☑ Processed (cleaned, filtered).
- ☑ Computed (metrics, deltas, confidence intervals).

⇒ the final step is to communicate the results via:

- Summaries: Tabular or statistical summaries of key metrics.
- Visualization: Graph that make effects easier to understand, especially for decision-makers.

> This step helps stakeholders quickly interpret whether the experiment succeeded, failed, or needs further investigation.

Real-World Analogy: Pizza A/B Test Finale

* You've cleaned and computed: *

- A group: 4.2 is average rating
- B group: 4.6 is with $p < 0.05$

Now you need to present results to your team or boss.

* Show summary tables of satisfaction scores *

- Show summary tables of satisfaction scores.
- Plot a bar chart of A vs B.
- Include confidence intervals.
- Maybe a time-series plot over days.

without this summary + visual:

> Your team can't make decisions based on evidence

② Key Components of Summary & Visualization.

1- Tabular Summary

A clean table showing:

- Mean metrics
- Delta
- p-values
- confidence intervals
- sample sizes.

Metric	A (control)	B (Treatment)	Delta	p-value	95% CI
Conversion Rate	12%	15%	+3%	0.01	(0.6%, 5.4%)

2. Visualization

Common plots:

- Bar plots with error bars.
- Time-series plots over days
- lift charts (relative improvements)
- funnel diagrams (for conversion steps)

Visuals make it easy to

- spot trends
- Interpret significance
- Detect outliers or anomalies.

(3) Math Concept Behind Visualization

Most visualizations show

- Point estimate (mean or proportion)
- Error margin (e.g. CI)

E.g. in a bar plot:

- Bar height = metric
- Error bar = $\pm$ 1.96 x SE (Standard Error)

This connects directly to Unit 6: Statistical Analysis whenever you learn:

$$CI \approx \bar{x} \pm 2 \cdot \frac{s}{\sqrt{n}}$$

Visuals are simply a graphical representation of this equation.

(4) Related / Overlapping Topics

Related Unit	Why it Matters
Unit 6 - Statistical Analysis	→ CI, SE, and p-values directly drive visual summaries.
Unit 9 - Metric score cards	→ Score cards often generate the summary tables.
Unit 10 - Trust & communication	→ Summary + visual = more trusted & fewer misleading results
Unit 21 - Automation	→ Summaries are often auto-generated in dashboards (e.g. in Airbnb's ERF, Microsoft's ExP platform).

(5) Summary

Practices		Description
Goal	⟶	Make experiments results interpretable.
Tools	⟶	Tables, bar charts, time-series, dot plots.
Core Math	⟶	CL, proportions, Std Error
Pit falls	⟶	Misleading charts, ignoring uncertainty bars.
Outcome	⟶	stakeholders make decisions quickly & confidently.

dhawa dorje ghising